plants.plt

Purpose

`plants.plt` is the plant growth database. Each row is one plant species, crop variety or land cover, with the parameters SWAT+ needs to grow it: plant type, heat units to maturity, leaf area development, radiation use efficiency, canopy height, rooting depth, temperature limits, nutrient content, stomatal conductance and residue properties.

Plant communities in `plant.ini`, plant operations in `management.sch` and decision tables, and land uses in `landuse.lum` all refer to rows of this file by name. The names are also used by the GIS interfaces to link land cover maps to SWAT+ land uses.

The file is named on the `hru_parm_db` line of `file.cio` (default `plants.plt`). If the file is missing, the database is empty.

The database distributed with SWAT+ has parameters for most common crops, grasses, trees and land covers. If a land use or plant is not in it, start from a row for a plant with similar growth habits.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one plant with 56 values in the order below. The reader does not match columns by header name, so every value must be present and in this order. It ignores anything after the 56th value, so the `description` column written by the SWAT+ Editor has no effect.

If the `event` column of `codes.bsn` is not 0, one more word is read after the 56th value: a plant class name of up to 25 characters (for example `row_crop`). In that case the plant class takes the place of the description.

#	Column	Units	Default	Description
1	`name`	-	-	Name of the plant or land cover, up to 40 characters. Must be unique.
2	`plnt_typ`	-	-	Plant type: `warm_annual`, `cold_annual`, `warm_annual_tuber`, `cold_annual_tuber` or `perennial`. See Plant types.
3	`gro_trig`	-	-	Growth trigger: `temp_gro` (growth driven by temperature and day length) or `moisture_gro` (tropical plants whose growth starts with the monsoon).
4	`nfix_co`	-	0.0	Nitrogen fixation coefficient: 0.5 for legumes, 0 for non-legumes. Any value above 0 lets the plant fix nitrogen when soil nitrate cannot meet its demand.
5	`days_mat`	days	110	Days to maturity, an integer. Used to compute the heat units to maturity. 0 means the plant uses the heat units of the whole year. See Heat units to maturity.
6	`bm_e`	(kg/ha)/(MJ/m²)	15.0	Biomass-energy ratio (radiation use efficiency): dry biomass produced, including roots, per unit of intercepted photosynthetically active radiation, with no stress, at ambient CO2 and a vapor pressure deficit of 1 kPa.

#	Column	Units	Default	Description
7	`harv_idx`	(kg/ha)/(kg/ha)	0.76	Harvest index for optimal growing conditions: fraction of above-ground biomass removed as yield. Values above 0.7 are replaced by 0.02 (see Values set by the model).
8	`lai_pot`	m^2/m^2	5.0	Maximum potential leaf area index. Values are limited to 0 to 10.
9	`frac_hu1`	fraction	0.05	Fraction of heat units to maturity at the 1st point of the optimal leaf area development curve.
10	`lai_max1`	fraction	0.05	Fraction of `lai_pot` at the 1st point of the optimal leaf area development curve.
11	`frac_hu2`	fraction	0.40	Fraction of heat units to maturity at the 2nd point of the optimal leaf area development curve.
12	`lai_max2`	fraction	0.95	Fraction of `lai_pot` at the 2nd point of the optimal leaf area development curve.
13	`hu_lai_decl`	fraction	0.99	Fraction of heat units to maturity at which leaf area starts to decline.
14	`dlai_rate`	-	1.0	Exponent of the leaf area decline of annuals after `hu_lai_decl`. 1 gives a linear decline.
15	`can_ht_max`	m	6.0	Maximum canopy height.
16	`rt_dp_max`	m	3.5	Maximum rooting depth. The soil profile depth also limits roots.
17	`tmp_opt`	°C	30.0	Optimal temperature for plant growth.
18	`tmp_base`	°C	10.0	Base (minimum) temperature for plant growth. Heat units accumulate only above it.
19	`frac_n_yld`	kg N/kg yield	0.0015	Fraction of nitrogen in the yield, dry weight.
20	`frac_p_yld`	kg P/kg yield	0.0003	Fraction of phosphorus in the yield, dry weight.
21	`frac_n_em`	kg N/kg biomass	0.006	Normal fraction of nitrogen in plant biomass at emergence.
22	`frac_n_50`	kg N/kg biomass	0.002	Normal fraction of nitrogen in plant biomass at 50% maturity.
23	`frac_n_mat`	kg N/kg biomass	0.0015	Normal fraction of nitrogen in plant biomass at maturity.
24	`frac_p_em`	kg P/kg biomass	0.0007	Normal fraction of phosphorus in plant biomass at emergence.
25	`frac_p_50`	kg P/kg biomass	0.0004	Normal fraction of phosphorus in plant biomass at 50% maturity.
26	`frac_p_mat`	kg P/kg biomass	0.0003	Normal fraction of phosphorus in plant biomass at maturity.
27	`harv_idx_ws`	(kg/ha)/(kg/ha)	0.01	Harvest index under severe water stress: the lowest harvest index expected, between 0 and `harv_idx`.
28	`usle_c_min`	-	0.001	Minimum USLE C factor for water erosion. Values are limited to 0.001 to 1.
29	`stcon_max`	m/s	0.002	Maximum stomatal conductance at high solar radiation and low vapor pressure deficit. Used with the Penman-Monteith ET method.
30	`vpd`	kPa	4.0	Vapor pressure deficit at the 2nd point of the stomatal conductance curve.
31	`frac_stcon`	fraction	0.75	Fraction of `stcon_max` at the 2nd point of the stomatal conductance curve.
32	`ru_vpd`	$(kg/ha)/(MJ/m^2)/kPa$	8.0	Rate of decline in radiation use efficiency per unit increase in vapor pressure deficit above 1 kPa. 0 turns the adjustment off.
33	`co2_hi`	µL CO2/L air	660.0	Elevated CO2 concentration at the 2nd point of the radiation use efficiency curve.

#	Column	Units	Default	Description
34	bm_e_hi	(kg/ha)/(MJ/m²)	16.0	Biomass-energy ratio at the CO2 concentration co2_hi.
35	plnt_decomp	-	0.05	Plant residue decomposition coefficient: fraction of residue that decomposes in a day under optimal moisture, temperature, C:N and C:P. 0 takes rsd_decomp from parameters.bsn.
36	lai_min	m²/m²	0.75	Minimum leaf area index of perennials during the dormant period.
37	bm_tree_acc	-	0.3	Exponent that limits the maximum leaf area index of young perennials. See Perennials and trees.
38	yrs_mat	years	10	Years for a perennial to reach full development, an integer. Values below 1 are set to 1.
39	bm_tree_max	t/ha	1000.0	Maximum above-ground biomass. Checked once a year.
40	ext_co	-	0.65	Light extinction coefficient for intercepted photosynthetically active radiation. 0 is replaced by 0.65.
41	leaf_tov_mn	months	12.0	Leaf turnover time of moisture_gro perennials with minimum stress. Called leaf_tov_min in earlier documentation.
42	leaf_tov_mx	months	3.0	Leaf turnover time of moisture_gro perennials with maximum stress. Called leaf_tov_max in earlier documentation.
43	bm_dieoff	fraction	0.0	Factor on the excess biomass removed when above-ground biomass exceeds bm_tree_max. 0 is replaced by 1.
44	rt_st_beg	fraction	0.0	Root to shoot ratio at the start of the growing season. 0 is replaced by 0.4.
45	rt_st_end	fraction	0.0	Root to shoot ratio at the end of the growing season. 0 is replaced by 0.2. Tuber types always use 0.7.
46	plnt_pop1	plants/m²	0.0	Plant population at the 1st point of the population and leaf area curve.
47	frac_lai1	fraction	0.0	Fraction of lai_pot at the 1st point of the population and leaf area curve.
48	plnt_pop2	plants/m²	0.0	Plant population at the 2nd point of the population and leaf area curve.
49	frac_lai2	fraction	0.0	Fraction of lai_pot at the 2nd point of the population and leaf area curve.
50	frac_sw_gro	-	0.5	Intended as the 30-day precipitation to PET ratio that starts growth of moisture_gro plants. Read but not used, which uses a fixed 0.5.
51	aeration	fraction	0.2	Aeration stress threshold: the fraction of pore space above field capacity that can fill with water before the plant is stressed by lack of air. 0 is replaced by 0.2.
52	rsd_pctcov	-	0.0	Residue factor for the percent cover equation. 0 is replaced by 0.4.
53	rsd_covfac	-	0.0	Residue factor for the surface cover (C factor) equation. 0 is replaced by 0.04.
54	avg_lig_frac	fraction	0.85	Average lignin fraction of residue. Read but not used.
55	ab_lig_frac	fraction	0.15	Lignin fraction of above-ground residue. Used by the dynamic carbon model only.
56	bg_lig_frac	fraction	0.12	Lignin fraction of below-ground residue (roots). Used by the dynamic carbon model only.

The defaults are the values SWAT+ uses for a component that is not set. They are not filled in for missing columns, since every row must have all 56 values.

Plant types

`plnt_typ` decides how the plant goes through the year.

- `warm_annual` and `cold_annual` are annual crops. A `cold_annual` is planted in the fall and goes dormant over winter when day length falls below the dormancy threshold. Leaf area declines after `hu_lai_decl`, following `dlai_rate`.
- `warm_annual_tuber` and `cold_annual_tuber` are annuals whose yield is below ground. Their root to shoot ratio at the end of the season is fixed at 0.7.
- `perennial` covers grasses, shrubs and trees. Perennials go dormant when day length falls below the threshold, keep at least `lai_min` of leaf area, and develop over `yrs_mat` years.

`gro_trig` = `moisture_gro` is for tropical plants. If the plant was planted with the monsoon option, it starts growing when the 30-day ratio of precipitation to potential ET reaches 0.5. Leaf turnover of `moisture_gro` perennials varies between `leaf_tov_mx` and `leaf_tov_mn` months with moisture stress.

Heat units to maturity

SWAT+ computes the heat units a plant needs to reach maturity at the start of the run, from the monthly mean temperatures of the HRU's weather generator station.

- `days_mat` = 0: the plant grows through the whole year. Heat units to maturity are 95% of the year's sum of mean daily temperature above `tmp_base`.
- `days_mat` > 0 for an annual: the planting date is taken as the day when the sum of mean daily temperatures above 0 °C reaches 15% of the annual sum. Heat units above `tmp_base` are then summed for `days_mat` days from that date.

The calculation works in both hemispheres. Giving the days of a variety (for example 90, 100, 110 or 120 days for corn) lets one database support several varieties. If `days_mat` is longer than the growing season at a site, the heat units of the whole season are used.

Leaf area development

The optimal leaf area index follows a curve through two points, (`frac_hu1`, `lai_max1`) and (`frac_hu2`, `lai_max2`), where the first value is the fraction of heat units to maturity and the second the fraction of `lai_pot`:

$$f_{LAI} = \frac{f_{PHU}}{f_{PHU} + \exp(l_1 - l_2\, f_{PHU})}$$

l_1 and l_2 are set at the start of the run so that the curve passes through both points. The values of `lai_pot` in the standard database are for typical plant densities in rainfed agriculture. Drought-prone regions with low planting densities, or irrigated land with high densities, may need different values. Canopy height grows with the square root of f_{LAI} up to `can_ht_max`.

When `plnt_pop1` and `plnt_pop2` are given, a second curve through (`plnt_pop1`, `frac_lai1`) and (`plnt_pop2`, `frac_lai2`) limits `lai_pot` by plant population.

Radiation use efficiency

When the `co2` column of `parameters.bsn` is above 350 µL/L, radiation use efficiency is adjusted for CO2 along a curve through (330 µL/L, `bm_e`) and (`co2_hi`, `bm_e_hi`). Otherwise `bm_e` is used as given. Values for `co2_hi` and `bm_e_hi` are needed even when climate change is not simulated, and `bm_e_hi` should be larger than `bm_e`. A `co2_hi` of 330 is replaced by 660.

When the vapor pressure deficit is above 1 kPa, radiation use efficiency drops by `ru_vpd` for each kPa above 1, down to no less than 27% of `bm_e`. A value of 6 to 8 is a reasonable approximation for most plants, and 0 turns the adjustment off.

Perennials and trees

The maximum leaf area index of a perennial grows with its age:

$$LAI_{max} = \text{lai_pot} \left(\frac{y}{\text{yrs_mat}} \right)^{\text{bm_tree_acc}}$$

where y is the number of years since the plant was established (at least 1), and LAI_{max} is at most `lai_pot`. With `bm_tree_acc` = 0 the plant reaches `lai_pot` in its first year.

At the end of each year, if above-ground biomass B exceeds `bm_tree_max` (converted to kg/ha), the amount

$$D = (1 + \text{bm_dieoff})\,(B - 1000\,\text{bm_tree_max})$$

dies and goes to the residue pools, with roots in the same proportion. Nothing dies if D is not smaller than B. A mature forest stand typically has 30 to 50 t/ha of biomass. `bm_dieoff` is not used at the start of dormancy in 62.0.1.

Nutrient uptake

The six fractions `frac_n_em` to `frac_p_mat` describe the nitrogen and phosphorus content of the whole plant, roots included, on a dry weight basis, at emergence, at 50% maturity and at maturity. They set the plant's nutrient demand through the season. Each fraction must be smaller than the one before it: if `frac_n_50` is not at least 0.0001 below `frac_n_em`, it is set to `frac_n_em` − 0.0001, and if `frac_n_mat` is not at least 0.0001 below `frac_n_50`, it is multiplied by 0.75. The phosphorus fractions are treated the same way.

Residue lignin

With `carbon` = 2 in `codes.bsn`, residue entering the soil is split into metabolic, structural and lignin pools. Below ground, the lignin fraction is `bg_lig_frac`, the structural fraction is `bg_lig_frac` / 0.8, and the rest is metabolic. Above ground the same split uses `ab_lig_frac`. With `carbon` = 0 these columns are not used.

Values set by the model

After the file is read, SWAT+ changes some values:

Column	Change
`harv_idx`	Values above 0.7 are set to 0.02.
`lai_pot`	Limited to 0 to 10.
`usle_c_min`	Limited to 0.001 to 1.
`plnt_decomp`	0 is replaced by `rsd_decomp` of `parameters.bsn`.
`yrs_mat`	Values below 1 are set to 1.
`ext_co`	0 is replaced by 0.65.
`bm_dieoff`	0 is replaced by 1.
`rt_st_beg`, `rt_st_end`	0 is replaced by 0.4 and 0.2. Tuber types get `rt_st_end` = 0.7.
`aeration`, `rsd_pctcov`, `rsd_covfac`	0 is replaced by 0.2, 0.4 and 0.04.
`co2_hi`	330 is replaced by 660.
`frac_n_50`, `frac_n_mat`, `frac_p_50`, `frac_p_mat`	Adjusted to decrease, as described above.

The `harv_idx` change affects crops whose harvest index is above 0.7, such as alfalfa (0.9 in the standard database) and tuber crops with a harvest index above 1. Their harvest index becomes 0.02.

Calibration

These parameters can be changed during calibration through `calibration.cal`: `bio_e` (`bm_e`), `hvsti` (`harv_idx`), `rdmx` (`rt_dp_max`), `t_opt` (`tmp_opt`), `t_base` (`tmp_base`) and `usle_c` (`usle_c_min`).

Example

The first rows of `refdata/Ames_sub1/plants.plt` in the SWAT+ repository:

```
plants.plt
name  plnt_typ  gro_trig  nfix_co  days_mat  bm_e  harv_idx  lai_pot  frac_hu1  lai_max1  frac_hu2  lai_ma
alfa  perennial  temp_gro  0.5  0  20  0.9  4  0.15  0.01  0.5  0.95  0.9  0.5  0.9  3  20  4  0.02  0.003
cana  warm_annual  temp_gro  0  100  34  0.3  4.5  0.15  0.02  0.45  0.95  0.5  0.3  1.3  1.4  21  5  0.03
```

- `alfa` (alfalfa) is a perennial legume (`nfix_co` = 0.5) with `days_mat` = 0, so it uses the heat units of the whole year. Its `harv_idx` of 0.9 is replaced by 0.02.
- `alfa` has `frac_n_50` = `frac_n_mat` = 0.02, so `frac_n_mat` is multiplied by 0.75 to 0.015.
- `bm_tree_max` is 0 for both rows. With `bm_dieoff` = 0.1, $D = 1.1\,B$ at the end of the year, which is not smaller than B, so no biomass dies off.
- Ames_sub1 has `event` = 0 and `carbon` = 2 in `codes.bsn`. No plant class is read, so the `description` column is ignored, and the lignin columns are used: for `alfa`, 9% of decomposing surface residue and 20% of root residue entering the soil goes to the lignin pool. `avg_lig_frac` is read but has no effect.

Related files

- `plant.ini`: plant communities for HRUs, by plant name.
- `management.sch` and decision tables: plant, harvest and kill operations name plants from this file.
- `landuse.lum`: land uses and their plant communities.
- `codes.bsn`: `event` switches the plant class column on; `carbon` selects whether the lignin fractions are used.
- `parameters.bsn`: `rsd_decomp` replaces a `plnt_decomp` of 0.
- Weather generator: monthly temperatures used for heat units to maturity.

Sources

- Reader: `plant_parm_read.f90`
- Type definition: `type plant_db` and `type residue_partition_fracs` in `plant_data_module.f90`
- Derived values and overrides: `plantparm_init.f90`
- Heat units to maturity: `plant_init.f90`
- Growth: `pl_leaf_gro.f90`, `pl_leaf_senes.f90`, `pl_root_gro.f90`, `pl_biomass_gro.f90`, `pl_dormant.f90`, `pl_mortality.f90`, `pl_nfix.f90`, `pl_waterup.f90`
- Monsoon trigger: `hru_control.f90`
- Residue lignin: `cbn_rsd_transfer.f90`, `cbn_surfrsd_decomp.f90`
- Calibration parameters: `cal_parm_select.f90`

Column	Fortran variable
`name`	`pldb%plantnm`
`plnt_typ`	`pldb%typ`
`gro_trig`	`pldb%trig`
`days_mat`	`pldb%days_mat`
`bm_e`	`pldb%bio_e`
`harv_idx`	`pldb%hvsti`
`lai_pot`	`pldb%blai`
`frac_hu1`, `lai_max1`	`pldb%frgrw1`, `pldb%laimx1`
`frac_hu2`, `lai_max2`	`pldb%frgrw2`, `pldb%laimx2`
`hu_lai_decl`	`pldb%dlai`
`can_ht_max`	`pldb%chtmx`
`rt_dp_max`	`pldb%rdmx`
`tmp_opt`, `tmp_base`	`pldb%t_opt`, `pldb%t_base`
`frac_n_yld`, `frac_p_yld`	`pldb%cnyld`, `pldb%cpyld`
`frac_n_em`, `frac_n_50`, `frac_n_mat`	`pldb%pltnfr1`, `pltnfr2`, `pltnfr3`
`frac_p_em`, `frac_p_50`, `frac_p_mat`	`pldb%pltpfr1`, `pltpfr2`, `pltpfr3`
`harv_idx_ws`	`pldb%wsyf`
`usle_c_min`	`pldb%usle_c`
`stcon_max`	`pldb%gsi`
`vpd`	`pldb%vpdfr`

Column	Fortran variable
frac_stcon	pldb%gmaxfr
ru_vpd	pldb%wavp
co2_hi, bm_e_hi	pldb%co2hi, pldb%bioehi
plnt_decomp	pldb%rsdco_pl
lai_min	pldb%alai_min
bm_tree_acc	pldb%laixco_tree
yrs_mat	pldb%mat_yrs
bm_tree_max	pldb%bmx_peren
ext_co	pldb%ext_coef
leaf_tov_mn, leaf_tov_mx	pldb%leaf_tov_min, pldb%leaf_tov_max
rt_st_beg, rt_st_end	pldb%rsr1, pldb%rsr2
plnt_pop1, frac_lai1	pldb%pop1, pldb%frlai1
plnt_pop2, frac_lai2	pldb%pop2, pldb%frlai2
frac_sw_gro	pldb%frsw_gro
avg_lig_frac	pldb%res_part_fracs%meta_frac
ab_lig_frac	pldb%res_part_fracs%str_frac
bg_lig_frac	pldb%res_part_fracs%lig_frac
plant class (event /= 0)	pl_class
event in codes.bsn	bsn_cc%nam1